9/15/2025

Data Visualization

Final Assignment

O. P. JINDAL UNIVERSITY

CALCULATED FIELDS AND TABLEAU WORKFLOW FOR SALES ANALYSIS

Data visualization tools like Tableau enable users to transform raw data into meaningful insights. Through features such as calculated fields and streamlined workflows, Tableau supports the exploration of patterns, trends, and relationships that may not be immediately evident. This discussion focuses on the process of creating calculated fields, their role in generating insights, and a complete Tableau workflow for analyzing sales and profitability across regions.

1 CREATING A CALCULATED FIELD IN TABLEAU

A calculated field in Tableau allows the creation of new data values by applying formulas or functions to existing fields. To create one, the user right-clicks in the *Data Pane*, selects **Create Calculated Field**, and enters a formula using dimensions, measures, or built-in functions. For example, to calculate profit margin, one might divide profit by sales and multiply by 100. Once created, the calculated field behaves like any other field and can be added to the visualization.

Calculated fields help uncover insights that are not visible in the raw dataset. For instance, raw data may contain total sales and profit, but a calculated profit margin provides a clearer understanding of efficiency across regions or time periods. This enhances decision-making by revealing relationships, trends, or ratios that would otherwise remain hidden (Murray, 2016).

2 TABLEAU WORKFLOW FOR SALES AND PROFITABILITY ANALYSIS

To analyze sales trends over time and profitability by region, Tableau provides a systematic workflow:

1. **Connecting to the Data Source**

   Begin by launching Tableau and connecting to the company's data source, such as an Excel file, SQL database, or cloud service. Verify the fields imported to ensure data completeness.

2. **Data Preparation**

   Review and clean the dataset by renaming fields, creating hierarchies, or applying filters. If needed, calculated fields like *Profit Margin* or *Sales Growth Rate* can be created at this stage for additional insights.

3. **Building Visualizations**
    - To track **sales trends over time**, place the *Date* field on the Columns shelf and *Sales* on the Rows shelf. Apply appropriate time granularity (e.g., monthly or quarterly). A line chart works well to illustrate growth or decline.
    - To examine **profitability by region**, place *Region* on the Columns shelf and *Profit* or *Profit Margin* on the Rows shelf. A bar chart or map can display differences clearly.
4. **Applying Filters and Formatting**
    Use filters to narrow the focus to specific periods or product categories. Formatting, color coding, and tooltips enhance readability and allow users to interact with the visualization.
5. **Combining into Dashboards**
    Combine multiple visualizations into a dashboard to provide a comprehensive view. A dashboard might include a sales trend line chart alongside a regional profitability bar chart, allowing users to compare metrics simultaneously.
6. **Interactivity**
    Add features like filters, drop-down menus, or highlight actions so decision-makers can explore data dynamically. This supports deeper investigation into specific regions or time frames.
7. **Publishing the Visualization**
    Once the dashboard is complete, publish it to Tableau Server, Tableau Cloud, or Tableau Public. This step ensures stakeholders across the company can access, share, and interact with the visualizations (Tableau, 2023).

Calculated fields expand the analytical power of Tableau by deriving new metrics from existing data. When combined with a structured workflow—from connecting to the data source through publishing dashboards—users can effectively analyze sales trends and profitability across regions. Together, these processes transform raw information into insights that support strategic decision-making.

3 EXPLANATION OF THE TREEMAP VISUALIZATION

I created a **Treemap** to analyze profit across different **Region–Category** combinations using the Sample Superstore dataset. The size of each tile represents the total profit for that combination, while the color intensity also corresponds to profit values. Each tile is labeled with the **Category, Region, Profit, and Profit as a Percentage of the Region's Total**.

From the visualization (attached at the end):

- The **highest profit** comes from **Office Supplies in the West region**, generating **$54,070**, which accounts for **42.9% of the total profit within the West region**.
- The **lowest profit** is observed in **Furniture in the East region**, generating only **$3,445**, which is **17.5% of the total profit within the East region**.

This treemap makes it easy to compare not just absolute profits but also how much each category contributes to the overall profitability within its region. By combining both measures (absolute profit and percentage of regional total), the visualization highlights which product categories are driving regional profitability and which are underperforming.

References

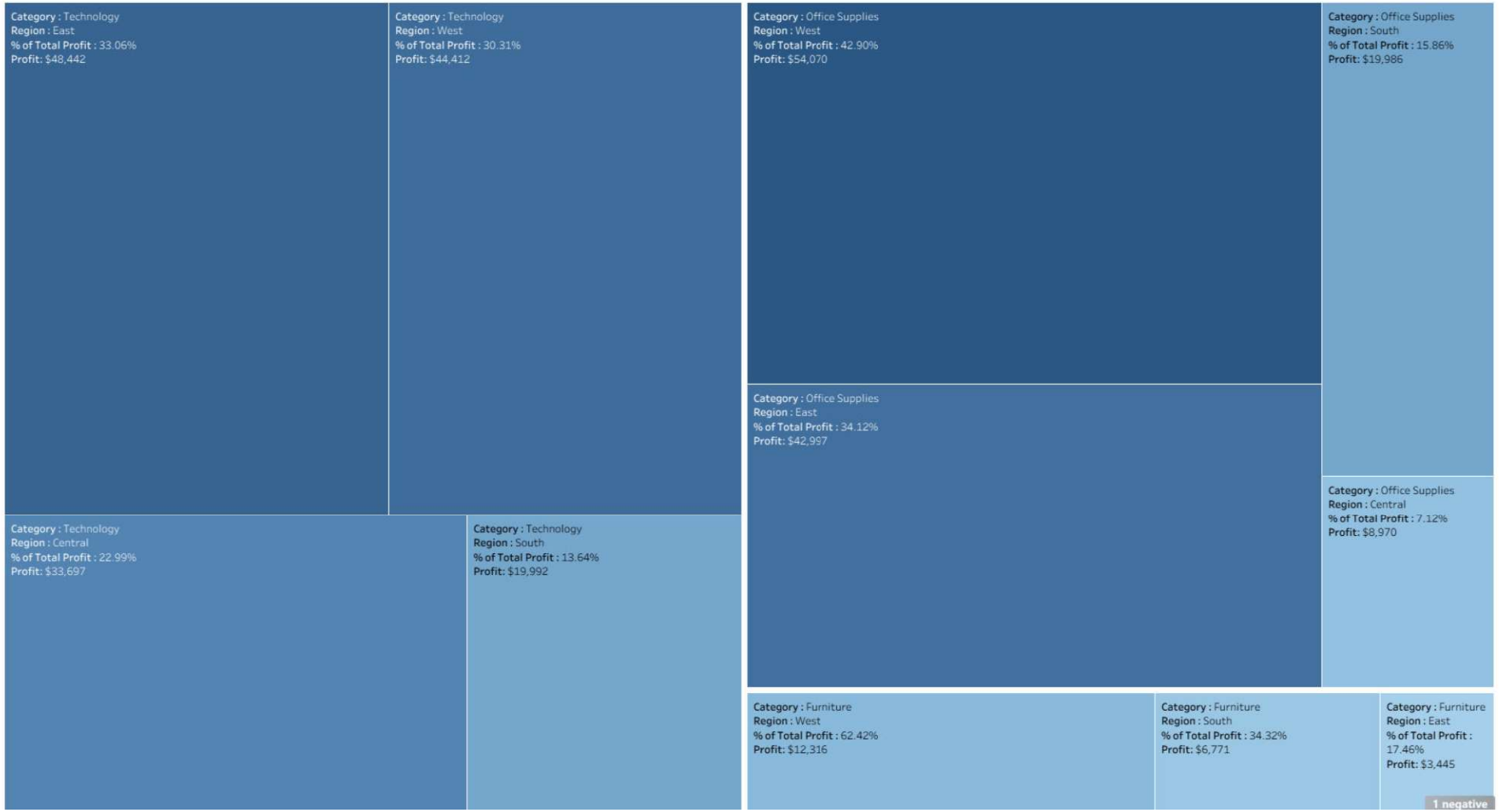
Murray, D. (2016). *Tableau your data!: Fast and easy visual analysis with Tableau software* (2nd ed.). Wiley. https://www.wiley.com/en-us/Tableau+Your+Data!%3A+Fast+and+Easy+Visual+Analysis+with+Tableau+Software%2C+2nd+Edition-p-9781119001195

Tableau. (2023). *Tableau desktop and web authoring help*. Retrieved from https://help.tableau.com

Wordcount: 730

Sheet 1

Most and Least Profitable Region–Category Combinations



Story | Layout

New story point

Blank | Duplicate

- Region_Map
- Region_Tree Map
- Region_Bar Chart
- Region_Time ...
- Region_Dashbo...
- Profit
- Positive_profit
- Negative_Profit
- Dashboard 2
- Cities ...
- Positive ...
- Negative ...
- Dashboard 3
- Time series plot
- Conclusion

A Drag to add text

Show title

Size

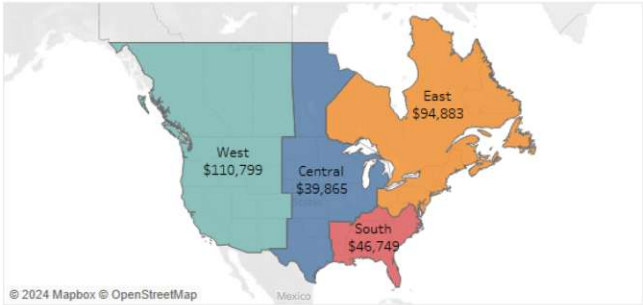
Generic Desktop (1366 x 7...

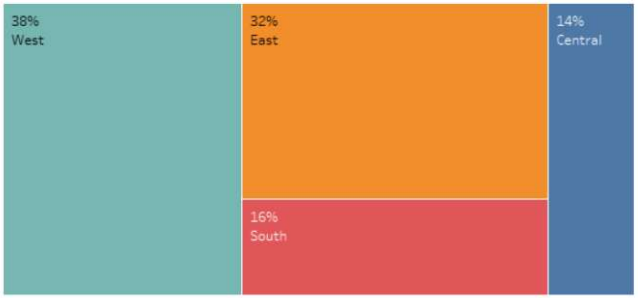
Story

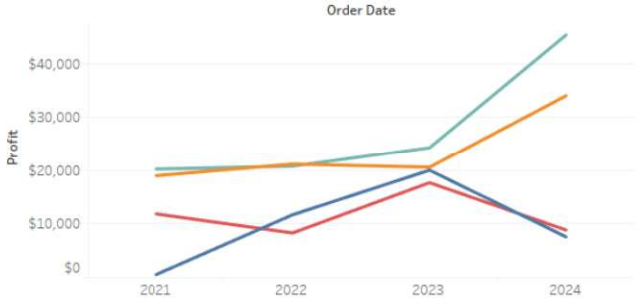
Overall Regional Profit Overview

How are the regions performing? | How are the states and citites are performing? | How significant are the city gains and losses? | How have the cities performing over time? | Conclusion

Regional profit overview







The West is the highest performing region, bringing in 38% of the overall total profit. The Southern and the Central regions, combined, are bringing in only 30% of the total profit.

Data Source | Region_Tree Map | Region_Bar Chart | Region_Time Series | Region_Dashboard | Profit | Positive_profit | Negative_Profit | Dashboard 2 | Citites Performance | Positive Performing Citites | Negative Performing Citites | Dashboard 3 | Time series plot | Conclusion | Story

Story | Layout

New story point

Blank | Duplicate

- Region_Map
- Region_Tree Map
- Region_Bar Chart
- Region_Time ...
- Region_Dashbo...
- Profit
- Positive_profit
- Negative_Profit
- Dashboard 2
- Citites ...
- Positive ...
- Negative ...
- Dashboard 3
- Time series plot
- Conclusion

A Drag to add text

Show title

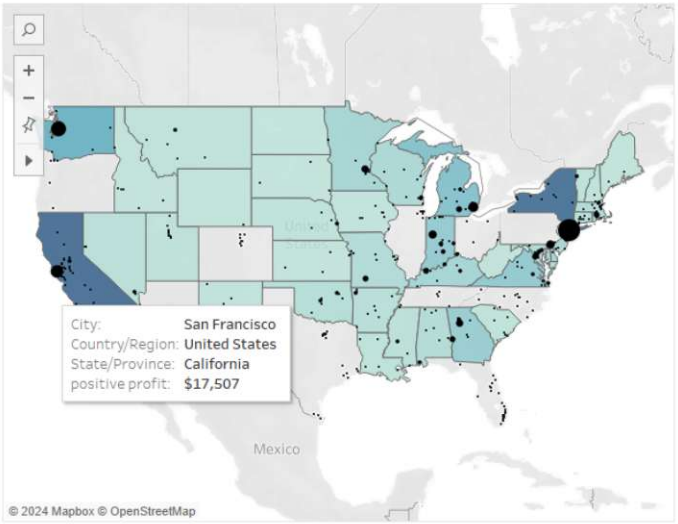
Size

Generic Desktop (1366 x 7...

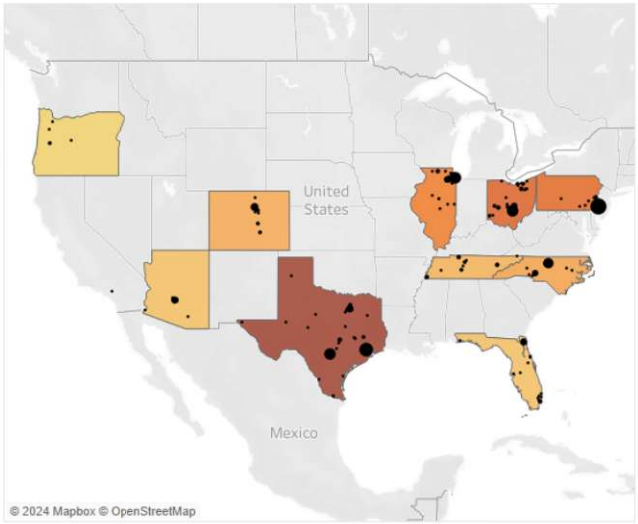
Story

How are the regions performing? | **How are the states and citites are performing?** | How significant are the city gains and losses? | How have the cities performing over time? | Conclusion

Profit Gain



Profit Loss



California and NewYork are the two most profitable states, both of which contain some of the most profitable citites. On the other hand Texas contains several of the least profitable citites, contributing to its overall status as the least profitable state.

positive profit: $0, $20,000, $40,000

positive profit: $100 – $76,381

negative profit: -$1, -$5,000, -$10,000

Profit: -$25,729 – -$1,190

Data Source | Region_Tree Map | Region_Bar Chart | Region_Time Series | Region_Dashboard | Profit | Positive_profit | Negative_Profit | Dashboard 2 | Citites Performance | Positive Performing Citites | Negative Performing Citites | Dashboard 3 | Time series plot | Conclusion | Story

Show Start Page (Ctrl+2)

New story point

Blank | Duplicate

- Region_Map
- Region_Tree Map
- Region_Bar Chart
- Region_Time ...
- Region_Dashbo...
- Profit
- Positive_profit
- Negative_Profit
- Dashboard 2
- Citites ...
- Positive ...
- Negative ...
- Dashboard 3
- Time series plot
- Conclusion

A Drag to add text

Show title

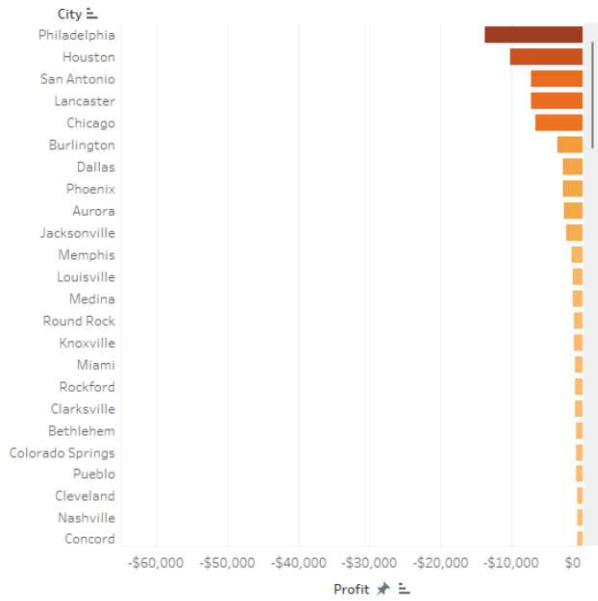
Size

Generic Desktop (1366 x 7...

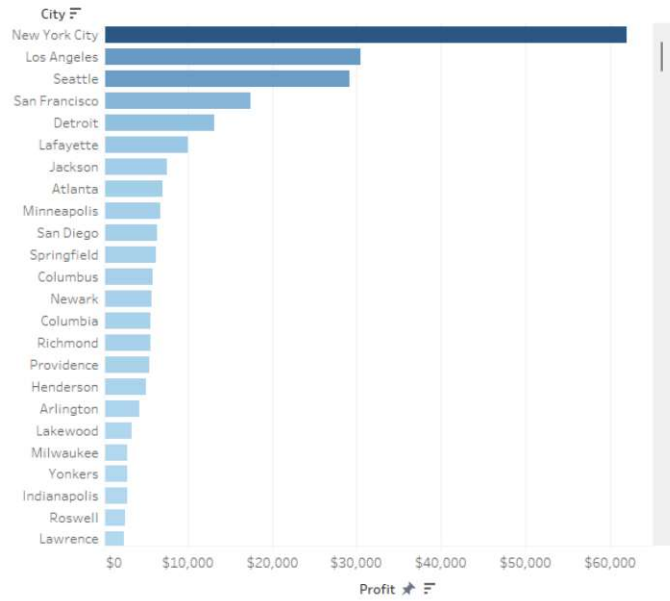
Story

How are the regions performing? | How are the states and citites are performing? | **How significant are the city gains and losses?** | How have the cities performing over time? | Conclusion

Profit Loss



Profit Gain



The extemities of the spectrum are particularly pronounced, with cities like NewYork, Los Angeles, and Seattle not just outperforming, but dominating the profit landscape, as indicated by their towering bars that dwarf others. On the other hands, cities like Philadelphia and Houston are not merely lagging, but is significantly anchoring the loss spectrum.

Data Source | Region_Tree Map | Region_Bar Chart | Region_Time Series | Region_Dashboard | Profit | Positive_profit | Negative_Profit | Dashboard 2 | Citites Performance | Positive Performing Citites | Negative Performing Citites | Dashboard 3 | Time series plot | Conclusion | Story

Story | Layout

New story point

Blank | Duplicate

- Region_Map
- Region_Tree Map
- Region_Bar Chart
- Region_Time ...
- Region_Dashbo...
- Profit
- Positive_profit
- Negative_Profit
- Dashboard 2
- Citites ...
- Positive ...
- Negative ...
- Dashboard 3
- Time series plot
- Conclusion

A Drag to add text

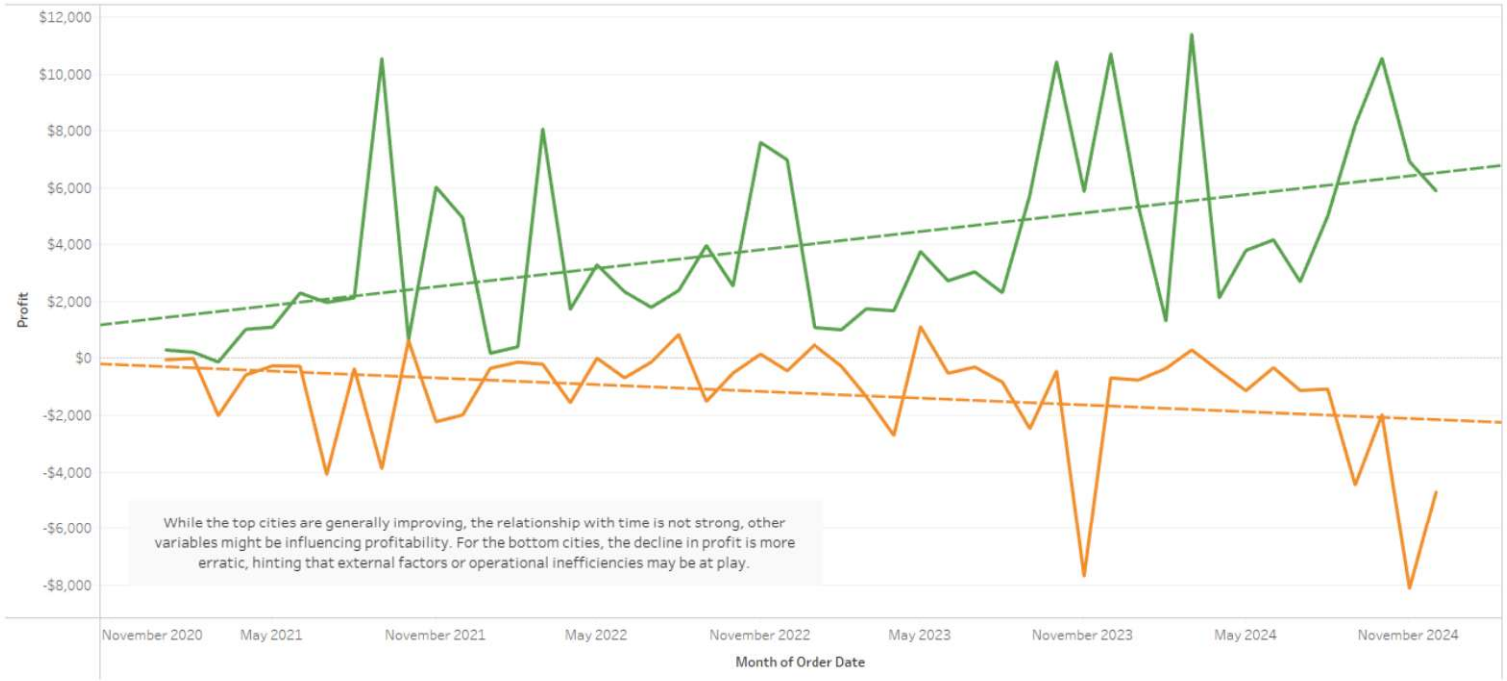
Show title

Size

Generic Desktop (1366 x 7...

Story

How are the regions performing? | How are the states and citites are performing? | How significant are the city gains and losses? | How have the cities performing over time? | Conclusion



While the top cities are generally improving, the relationship with time is not strong, other variables might be influencing profitability. For the bottom cities, the decline in profit is more erratic, hinting that external factors or operational inefficiencies may be at play.

Data Source | Region_Tree Map | Region_Bar Chart | Region_Time Series | Region_Dashboard | Profit | Positive_profit | Negative_Profit | Dashboard 2 | Citites Performance | Positive Performing Citites | Negative Performing Citites | Dashboard 3 | Time series plot | Conclusion | Story

Story | Layout

New story point

Blank | Duplicate

- Region_Map
- Region_Tree Map
- Region_Bar Chart
- Region_Time ...
- Region_Dashbo...
- Profit
- Positive_profit
- Negative_Profit
- Dashboard 2
- Citites ...
- Positive ...
- Negative ...
- Dashboard 3
- Time series plot
- Conclusion

Drag to add text

Show title

Size

Generic Desktop (1366 x 7...

Story

How are the regions performing? | How are the states and citites are performing? | How significant are the city gains and losses? | How have the cities performing over time? | Conclusion

The Western and Eastern regions have been more profitable than the Central and Southern regions. Given that a small number of cities are driving both profit and loss, resources and efforts should be prioritized on the top performing cities to maximize profitability. Simultaneously, targeted strategies to mitigate losses in the lowest performing cities should be implemented.

While there is some trend over time with the top and bottom performing cities, other factors likely have a more significant impact on the profit trends for these cities.

For the top performing cities, focus on reinforcing and expanding successful strategies, such as investing in high performing product categories or optimizing marketing efforts.

For the underperforming cities, conduct a deep dive analysis to identify root causes of losses, such as supply chain inefficiencies, suboptimal pricing strategies, or local competition. Implement corrective measures tailored to each city's specific challenges.

Additionally, leverage learnings from top cities to potentially replicate their success in the bottom cities. This bifocal approach, capitalizing on strengths in profitable areas while addressing weaknesses in underperforming ones - will help balance overall profitability across regions.

Data Source | Region_Tree Map | Region_Bar Chart | Region_Time Series | Region_Dashboard | Profit | Positive_profit | Negative_Profit | Dashboard 2 | Citites Performance | Positive Performing Citites | Negative Performing Citites | Dashboard 3 | Time series plot | Conclusion | Story